NARASARAOPETA ENGINEERING COLLEGE (AUTONOMOUS)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

2025 - 2026

Batch Number	BG-1
Team Members	Chinnapareddy Anjali Reddy(22471A0583) Dogiparthi Dhanusha (22471A0587) Shaik Chikli Asifa(22471A0581)
Guide	**Dr. Sireesha Moturi** B.Tech., M.Tech., Ph.D Associate Professor
Title	**Exploring Quantum Machine Learning for Enhanced Skin Lesion Classification: A Comparative Study of Implementation Methods**
Domain/Technology	DEEP LEARNING
Base Paper Link	**https://ieeexplore.ieee.org/document/10613907**
Dataset Link	**https://www.kaggle.com/datasets/kmader/skin-cancer-mnist-ham10000**
Software Requirements	Browser: Any latest browser like Chrome Operating System: Windows 7 Server or later Python (COLAB)
Hardware Requirements	SystemType: Intel Core i5 or above RAM: 8 GB Number of cores:5 Number of Threads: 4
Abstract	Skin lesion classification is important for early medical diagnosis and treatment. In our previous work, quantum machine learning models were studied, but their performance was limited due to simulator constraints and low accuracy. To address these limitations, this work implements classical deep learning models along with quantum models for comparison. EfficientNet-B0, MobileNetV3, ResNet18, and EFAM-Net were trained using the same dataset and preprocessing methods. All models were evaluated under identical experimental conditions to ensure a fair comparison. The results show that classical deep learning models achieve higher accuracy and better training stability than quantum-based approaches. This study concludes that classical methods are currently more suitable for real-world skin lesion classification.

Signature of the student(s) **Signature of the Guide** **Signature of the project coordinator**